

The empirical recurrence for OEIS A251290, recovered from its tail

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Abstract

OEIS A251290 records “Empirical recurrence of order 58” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 128$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 60$. Its last coefficient is $c_{58} = -64 \neq 0$, so no further tail satisfies anything shorter; any order-58 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A251290 is “Number of $(n+1) \times (6+1)$ 0..1 arrays with every 2×2 subblock having a single 1 or two 1s on the same edge or main diagonally.”. It has offset 1 and begins

710, 6980, 52474, 518711, 4109933, 39473392, 322933003, 3029148345, ...

The entry states:

Empirical recurrence of order 58 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:14:43, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 128$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 128$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 58: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 58.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 256$ exact terms from $n = 2$ on, it returns order 58 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{58} c_i a(n-i),$$

c_1	2	c_{16}	−676138190339	c_{31}	1135827010847627	c_{46}	−2916776167495
c_2	195	c_{17}	−139561663286	c_{32}	−2357779413848241	c_{47}	1595379431600
c_3	−285	c_{18}	4449928188387	c_{33}	−1455932437072218	c_{48}	455675581751
c_4	−15919	c_{19}	1343993695496	c_{34}	2084187785184164	c_{49}	−139948503367
c_5	17097	c_{20}	−22721863705089	c_{35}	1456292979102740	c_{50}	−41954574890
c_6	738366	c_{21}	−8428166786770	c_{36}	−1358277313428806	c_{51}	9037121662
c_7	−563046	c_{22}	90639637138053	c_{37}	−1115573314865193	c_{52}	2815902234
c_8	−22161306	c_{23}	38029573216251	c_{38}	616302281450840	c_{53}	−114928835
c_9	11062246	c_{24}	−283694310569114	c_{39}	636667788938783	c_{54}	−63643416
c_{10}	462269714	c_{25}	−129205133907504	c_{40}	−169858062365402	c_{55}	640460
c_{11}	−126215601	c_{26}	697697832762711	c_{41}	−260891589901767	c_{56}	447348
c_{12}	−6996887855	c_{27}	339193943876402	c_{42}	14131537274376	c_{57}	−8704
c_{13}	561431670	c_{28}	−1346001348237567	c_{43}	73285791523932	c_{58}	−64
c_{14}	78955109193	c_{29}	−698509769376649	c_{44}	7567902329071		
c_{15}	6286416593	c_{30}	2026468562943417	c_{45}	−13441496919461		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 60$, and it is the recurrence OEIS A251290 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{58} - c_1 x^{57} - \dots - c_{58}$. Its constant term is $-c_{58} = 64$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 58 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 58, so $\deg q = 58$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 58, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 58, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -64 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-58 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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